Subject: Engineering Mathematics III **Paper Code: BEM-C302** **Maximum Marks: 20**

Section A (Short Answer type Questions)

Attempt any two questions from this section. Each question carries 6 marks.

Q1. Evaluate Laplace transform of $\int_0^t e^{-2t}\, t^3 dt$.

Q2. Express the following function in terms of Heaviside unit step function and hence find the Laplace transform $f(t) = \begin{cases} t^2 & 0 < t < 1 \\ 4t & t > 1 \end{cases}$

Q3. Apply convolution theorem to find inverse Laplace transform of $\frac{1}{(s-2)(s+2)^2}$

Q4. Find Laplace transform of $\frac{e^{-t} sint}{t}$

Section B (Long Answer type Questions)

Attempt any one questions from this section. Each question carries 8 marks.

Q1. Apply Laplace transform to solve $y'' + 4y' + 8y = 1, y(0) = 0, y'(0) = 1.$

Q2. Determine the response of the damped mass spring system governed by

$$y'' + 3y' + 2y = r(t), \quad y(0) = 0, \quad y'(0) = 0$$

Where r(t) is the unit impulse at time $t = 1, r(t) = \delta(t-1)$.

I Sessional B.Tech. III Sem. (ECE & EE) BEM - C302

Sec. A (Attempt any two questions) M.M. - 6 × 2	Sec. B (Attempt any one question) M.M. - 8 × 1
1. Find the values of C_1 and C_2 such that the function $f(z)= x^2 + C_1y^2 - 2xy + i(C_2x^2 - y^2 + 2xy)$ is analytic. Also find $f'(z)$ 2. Use Cauchy's Integral formula to evaluate $\int_C \frac{z}{z^2-3z+2}dz$, where C is the circle $\lvert z - 2\rvert = \frac{1}{2}$. 3. Evaluate $\int_C \lvert z\rvert dz$ where c is the left half of the unit circle $\lvert z\rvert = 1$ from z = -i to z = i. 4. Evaluate the following integral using residue theorem $\int_C \frac{4-3z}{z(z-1)(z-2)}dz$, where C is the circle $\lvert z\rvert = \frac{3}{2}$	1. Use residue calculus to evaluate the following integral $\int_0^{2\pi} \frac{1}{5-4\sin\theta}d\theta$. 2. Prove that $u=3x - 2xy$ is harmonic. Find a conjugate function v such that $f(z)=u + iv$ is analytic. Also express f(z) in terms of z.

Engineering Mathematics III BEM-C302 I Sessional CSE & EE

Section A(Attempt any two questions. Each question carries 6 marks)

Q1. Find Laplace transform of $e^{-4t}\int_0^t \boxed{t\sin^2 t}\,dt$ → $\frac{1 - \cos 2t}{2}$

Q2. Using Laplace transform evaluate $\int_0^\infty \left(\frac{e^{-t}-e^{-2t}}{t}\right)dt$. → $\log\left(\frac{s+2}{s+1}\right)$

Q3. Find inverse Laplace transform of $\frac{1}{(s-2)(s+2)^2}$

Q4. Find the Fourier integral representation of the function $f(x) = \begin{cases} e^{ax} & x \le 0 \\ e^{-ax} & x \ge 0 \end{cases}$ for $a > 0$

Hence show that $\int_0^\infty \frac{\cos\lambda x}{\lambda^2+a^2}d\lambda = \frac{\pi}{2a}e^{-ax}$, $x \ge 0$.

Section B(Attempt any one question. Each question carries 8 marks)

Q1. Express the function $f(x) = \begin{cases} 1 & when\ |x| \le 1 \\ 0 & when\ |x| > 1 \end{cases}$ as Fourier integral. Hence, evaluate $\int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}d\lambda$.

Q2. Use Laplace transform to solve the differential equation $y'' - 3y' + 2y = 4$, $y(0)=2$, $y'(0)=3$.

Engineering Mathematics III BEM-C302 II Sessional **CSE, EE, ME**

Section A (Attempt any two questions. Each question carries 6 marks)

Q1. Show that Z transform of $\frac{1}{n!} = e^{\frac{1}{z}}$ hence evaluate Z transform of $\frac{1}{(n+1)!}$ and $\frac{1}{(n+2)!}$.

Q2. Find Z transform of $(i)\ n\sin n\theta$ $\quad (ii)\ a^n n^2$

Q3. Use iteration method to approximate the real root of $\cos x = 3x - 1$, correct to 3 decimal places if the root $\in (0,1)$.

Q4. Use Newton Raphson method to approximate the real root of $x^5 - 5x^2 + 3 = 0$, correct to 4 decimal places.

Section B (Attempt any one question. Each question carries 8 marks)

Q1. Use Z transform to solve $y_{n+2} + 4y_{n+1} + 3y_n = 3^n$ given $y_0 = 0, y_1 = 1$.

Q2. Use Bisection method to find a real root of the equation $x^3 - 4x - 9 = 0$ correct to 3 decimal places if the root $\in (2,3)$.